Screen Time & Mental Wellness — Data Analytics Case Study

1. **Project Objective**

This project investigates the relationship between screen time, mental wellness, and productivity.

2. **Primary Hypothesis**

Higher screen time, particularly leisure screen time, is associated with lower mental wellness and productivity.

3. **Secondary Research Question**

Among people with higher screen time, are exercise and social interaction associated with better mental wellness?

4. **Data Validation and Quality Checks**

Dataset Overview

- Number of records: **400**
- User IDs: **All unique**
- Missing values: **None**
- Out-of-range values: **None**
- Screen-time consistency check: **Passed**
- Formula checked: **Total Screen Time - Work Screen Time - Leisure Screen Time = 0**

Result:

The dataset contains 400 records and passed the initial data-quality checks. All user-IDs are unique, there are no missing values, and no out-of-range values, and total screen time is consistent with the sum of work screen time and leisure screen time.

5. Descriptive Statistics

Variable	Mean	Median	Std Dev	Min	Max
Total Screen Time	9.0249	9.09	2.4911	1.00	19.17
Work Screen Time	2.1831	1.455	1.9313	0.11	12.04
Leisure Screen Time	6.8418	6.70	2.2209	0.89	13.35
Mental Wellness	20.3268	14.80	20.3768	0	97
Productivity	54.3065	51.75	15.0201	20.6	100
Stress	8.1505	8.80	2.0948	0	10
Exercise Minutes	109.4525	103	70.2020	0	372
Social Hours	7.9050	7.75	4.9096	0	23.9
Sleep Hours	7.0132	7.03	0.8524	4.64	9.74
Sleep Quality	1.3975	1	0.6523	1	4

Key Observations

- Average total screen time was approximately 9.02 hours, of which approximately 2.18 hours were work-related and 6.84 hours were leisure-related.
- Mental wellness had a mean of 20.33 and a median of 14.8, indicating substantial variation in wellness scores.
- Average productivity was 54.31.
- Average exercise was approximately 109 minutes per week, while social interaction averaged 7.91 hours per week.
- Average sleep was approximately 7.01 hours per night.

6. **Correlation Analysis**

Correlation Analysis	
Relationship	**Correlation**
Total Screen Time ↔ Mental Wellness	-0.6359429863
Leisure Screen Time ↔ Mental Wellness	-0.4643660099
Work Screen Time ↔ Mental Wellness	-0.2862613961
Total Screen Time ↔ Productivity	-0.706537393
Leisure Screen Time ↔ Productivity	-0.50055769
Work Screen Time ↔ Productivity	-0.3356974299
Exercise → Mental Wellness	0.2330500626
Social Hours → Mental Wellness	0.07039977841
Sleep Hours → Mental Wellness	0.5808239958
Sleep Quality → Mental Wellness	0.750092297
Stress → Mental Wellness	-0.9138180536

Key Findings

Total screen time showed a moderately strong negative association with mental wellness (r = -0.636) and a strong negative association with productivity (r = -0.707).

Leisure screen time also showed negative associations with mental wellness (r = -0.464) and productivity (r = -0.501). Work screen time showed weaker negative associations with mental wellness (r = -0.286) and productivity (r = -0.336).

Overall, the results indicate that higher screen time is associated with lower mental-wellness and productivity scores in this dataset. Leisure screen time showed stronger negative associations with both outcomes than work screen time.

These correlations indicate associations and do not establish that higher screen time causes lower mental wellness or productivity.

7. High vs Low Leisure Screen Time Analysis

To further investigate the relationship between leisure screen time and mental wellness and productivity, participants were divided into two groups using the median leisure screen time of 6.7 hours per day as the cutoff. Participants with leisure screen time of 6.7 hours or less were classified as the low-leisure-screen group, while participants above 6.7 hours were classified as the high-leisure-screen group.

Measure	Low Leisure Screen (≤6.7h)	High Leisure Screen (>6.7h)
Participants	201	199
Mental Wellness	27.61	12.97
Productivity	59.96	48.59

The low-leisure-screen group had substantially higher average mental-wellness and productivity scores than the high-leisure-screen group. Average mental wellness was 27.61 in the low-screen group compared with 12.97 in the high-screen group. Average productivity was 59.96 compared with 48.59, respectively.

These group differences are consistent with the negative correlations observed between leisure screen time, mental wellness, and productivity. However, the comparison does not establish that reducing leisure screen time directly causes improvements in these outcomes.

8. Secondary Research Question Analysis

8.1 High Screen Time Group

To investigate whether exercise and social interaction were associated with better mental wellness among participants with higher screen time, the 400 participants were divided into low- and high-total-screen-time groups using the median total screen time of 9.09 hours per day. Participants with total screen time of 9.09 hours or less were classified as the low-screen group, while participants with more than 9.09 hours were classified as the high-screen group. Each group contained 200 participants.

8.2 Exercise and Mental Wellness

Exercise Group - High screen participants	Participants Count	Average mental wellness
Lower Exercise (<=103 minutes)	116	9.022413793
Higher Exercise (>103 minutes)	84	11.00238095

Among participants with higher total screen time, the higher-exercise group had a higher average mental-wellness score (11.00) than the lower-exercise group (9.02). However, the correlation between exercise and mental wellness within the high-screen group was very weak and positive ($r = 0.098$). Therefore, the data provides only limited evidence of a meaningful linear association between exercise and mental wellness among high-screen-time participants

8.3 Social Interaction and Mental Wellness

Social Group - High Screen Participants	Participants Count	Average Mental Wellness
Lower Social Hours (<=7.75 hours)	117	10.8982906
Higher Social hours(>7.75 hours)	83	8.381927711

Among participants with higher total screen time, the higher-social group had a lower average mental-wellness score (8.38) than the lower-social group (10.90). The correlation between social hours and mental wellness within the high-screen group was also very weak and negative ($r = -0.121$). Therefore, this dataset does not provide evidence that greater social hours are associated with better mental wellness among high-screen-time participants.

8.4 Answer to the Secondary Research Question

The analysis does not strongly support the secondary research question. Among participants with higher screen time, exercise showed a very weak positive association with mental wellness, while social hours showed a very weak negative association. Although participants with higher exercise had a somewhat higher average mental-wellness score, the correlation was close to zero. The data therefore does not provide strong evidence that exercise or greater social interaction is associated with better mental wellness within the high-screen-time group.

These findings describe associations within this dataset and do not establish cause and effect. Other factors, such as stress, sleep, or other characteristics of the participants, may also be related to mental wellness.

9. Additional Factors Associated with Mental Wellness

Factor	Correlation with Mental Wellness
Sleep Hours	0.581
Sleep Quality	0.75
Stress	-0.914

Additional analysis examined other factors associated with mental wellness. Sleep hours showed a moderate positive association with mental wellness ($r = 0.581$), while sleep quality showed a strong positive association ($r = 0.750$). Stress showed a very strong negative association with mental wellness ($r = -0.914$).

The linear model between stress and mental wellness produced an R^2 of 0.835, indicating a strong linear association between the two variables in this dataset. This result suggests that stress is an important factor associated with variation in mental-wellness scores. However, the analysis is observational and does not establish a causal relationship.

10. Key Findings

Finding 1 — Screen time and mental wellness

Higher total screen time was associated with lower mental-wellness scores ($r = -0.636$). Leisure screen time also showed a moderate negative association with mental wellness ($r = -0.464$), while work screen time showed a weaker negative association ($r = -0.286$).

Finding 2 — Screen time and productivity

Total screen time showed a strong negative association with productivity (r = -0.707). Leisure screen time also showed a moderate negative association with productivity (r = -0.501), while work screen time showed a weaker negative association (r = -0.336).

Finding 3 — Low vs high leisure screen time

Participants in the low-leisure-screen group had substantially higher average mental-wellness and productivity scores than participants in the high-leisure-screen group. Average mental wellness was 27.61 compared with 12.97, while average productivity was 59.96 compared with 48.59.

Finding 4 — Exercise and social interaction

Among high-screen-time participants, exercise showed only a very weak positive association with mental wellness (r = 0.098), while social hours showed a very weak negative association (r = -0.121). Therefore, the analysis does not provide strong evidence that exercise or greater social interaction is associated with better mental wellness within this subgroup.

Finding 5 — Sleep and stress

Sleep hours and sleep quality were positively associated with mental wellness (r = 0.581 and r = 0.750, respectively), while stress showed a very strong negative association with mental wellness (r = -0.914). The stress-versus-mental-wellness linear model had an R^2 of 0.835.

11. Conclusion

The analysis provides evidence supporting the primary hypothesis that higher screen time is associated with lower mental wellness and productivity in this dataset. Total screen time showed negative associations with both mental wellness and productivity, with the relationship with productivity being particularly strong. Leisure screen time also showed negative associations with both outcomes and had stronger relationships than work screen time.

The high- versus low-leisure-screen comparison further showed that participants with lower leisure screen time had substantially higher average mental-wellness and productivity scores. However, these findings represent associations rather than causal effects, so the analysis cannot establish that reducing screen time would directly improve mental wellness or productivity.

The secondary research question was not strongly supported. Among high-screen-time participants, exercise had only a very weak positive association with mental wellness, while social hours had a very weak negative association. In contrast, sleep and stress showed stronger associations with mental wellness, particularly stress.

Overall, the findings suggest that leisure screen time is an important factor to consider when examining mental wellness and productivity, but it should be considered alongside other factors such as stress and sleep.

12. Recommendations

1. Monitor leisure screen time

Since higher leisure screen time was consistently associated with lower mental wellness and productivity, individuals and organizations could consider monitoring leisure screen use and identifying opportunities to reduce unnecessary screen exposure.

2. Pay attention to stress

Stress showed the strongest negative association with mental wellness in this dataset. Stress management should therefore be considered alongside screen-time habits when thinking about mental wellness.

3. Support healthy sleep habits

Sleep hours and especially sleep quality showed positive associations with mental wellness. Maintaining adequate and consistent sleep may be an important area to consider when evaluating factors related to mental wellness.

4. Don't assume exercise or social interaction alone will solve the problem

The subgroup analysis did not show strong evidence that exercise or social hours were positively associated with mental wellness among high-screen-time participants. Further analysis would be needed before recommending these factors as specific interventions based on this dataset alone.

5. Distinguish work and leisure screen use

Work screen time showed weaker negative associations with mental wellness and productivity than leisure screen time. This suggests that future analysis could distinguish between necessary work-related screen exposure and discretionary leisure screen use rather than treating all screen time identically.

13. Limitations

1. Observational data

This analysis identifies relationships and patterns within the dataset but does not establish causation. For example, the negative association between screen time and mental wellness does not prove that higher screen time directly causes lower mental wellness.

2. Correlation does not account for all factors

Correlation analysis examines relationships between pairs of variables and does not account for all other factors that may influence the outcome. Variables such as stress, sleep, and other characteristics may contribute to differences in mental wellness.

3. Group comparisons use median-based cutoffs

The low- and high-screen groups were created using median values as practical cutoffs. These groups simplify continuous variables and may hide differences among individuals within each group.

4. Results are specific to this dataset

The findings are based on 400 records in this dataset and may not generalize to other populations, age groups, occupations, or environments.

5. No causal intervention was tested

The analysis does not test what happens when participants actually reduce screen time, increase exercise, increase social interaction, or change their sleep habits. Therefore, the recommendations should be viewed as areas for consideration rather than proven interventions.